

Technology Stack

Date	02 July 2026
Team ID	XXXXXX
Project Name	Smart Lender - AI Powered Loan Eligibility Prediction
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, libraries, and tools used to build the Smart Lender solution. A well-chosen technology stack ensures that the application is scalable, maintainable, and aligned with the project requirements.

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3, Bootstrap 5, Font Awesome 6, Google Font (Poppins)	Responsive, modern glassmorphism-style UI with minimal custom CSS
2	Backend / Server-Side	Python 3.12, Flask 3.1	Lightweight framework well suited for serving ML models
3	Machine Learning	Scikit-learn (KNN, Decision Tree, Random Forest), XGBoost	Compare multiple classifiers and deploy the best performing one
4	Data Handling	Pandas, NumPy	Data loading, cleaning, encoding and manipulation
5	Model Persistence	Joblib	Save/load the trained model, scaler and label encoders (.pkl files)
6	Visualization	Matplotlib	Exploratory data analysis and evaluation charts (confusion matrix, etc.)
7	WSGI / Deployment	Gunicorn, Procfile, runtime.txt	Production-grade serving on cloud platforms (e.g., Render/Heroku)
8	Version Control & CI/CD	Git & GitHub	Code management, versioning and collaboration